

Duty 13309 — chronological leg key

L = passenger-leg order; M = inter-area empty-move order. Trip numbers are stable prepared-input labels, not GIRO-supplied journey numbers. The graph groups platforms through documented reference areas; the codes below retain actual recorded endpoints.

Step	Graph	Trip	Recorded endpoints	Departure	Arrival	Minutes
1	M1		PARX → HED_D	05:19	05:37	18
2	L1	226	HED_D → PC_10	05:37	06:07	30
3	L2	229	PC_10 → HED_E	06:07	06:41	34
4	L3	234	HED_D → PC_10	06:57	07:27	30
5	L4	586	PC_7 → GMV_G	07:30	07:54	24
6	L5	589	GMV_H → PC_7	08:00	08:27	27
7	L6	901	PC_9 → JON_B	08:27	08:42	15
8	L7	792	JON_A → PC_7	08:45	08:57	12
9	L8	793	PC_7 → JON_B	08:57	09:14	17
10	L9	796	JON_A → PC_7	09:25	09:37	12
11	L10	393	PC_5 → HED_E	09:37	10:08	31
12	L11	396	HED_D → PC_5	10:13	10:37	24
13	M2		PC_5 → PARX	10:37	10:42	5
14	M3		PARX → JON_A	12:43	12:54	11
15	L12	922	JON_A → PC_9	12:54	13:07	13
16	L13	626	PC_7 → GMV_G	13:10	13:34	24
17	L14	629	GMV_H → PC_7	13:42	14:07	25
18	L15	632	PC_7 → GMV_G	14:10	14:35	25
19	L16	635	GMV_H → PC_7	14:42	15:07	25
20	L17	638	PC_7 → GMV_G	15:10	15:35	25
21	L18	641	GMV_H → PC_7	15:40	16:07	27
22	M4		PC_7 → 3127L	16:07	16:22	15
23	L19	447	HED_D → PC_5	16:50	17:17	27
24	L20	451	PC_5 → HED_E	17:17	17:48	31
25	L21	456	HED_D → PC_5	17:52	18:17	25
26	L22	459	PC_5 → HED_E	18:17	18:47	30
27	M5		HED_E → PARX	18:47	19:02	15

Duty 13309 — numbered area visits

Visit bounds include local platform moves and dwells. The full event ledger follows. V numbering counts repeated arrivals to an area; it is separate from L/M/C numbering.

Visit	Area / recorded platform codes	Arrival	Departure	Charges / visit detail
V1	PARX depot · PARX	05:12	05:19	prep PARX 05:12–05:19
V2	Heden · HED_D	05:37	05:37	Immediate departure
V3	Partille centrum · PC_10	06:07	06:07	Immediate departure
V4	Heden · HED_E, 3127L, HED_D	06:41	06:57	deadhead HED_E→3127L 06:41–06:42; C1 charge 3127L 06:42–06:56 +40.594kWh; deadhead 3127L→HED_D 06:56–06:57
V5	Partille centrum · PC_10, PC_7	07:27	07:30	same reference gap PC_10→PC_7 07:27–07:30
V6	Gamlestads Torg · GMV_G, GMV_H	07:54	08:00	deadhead GMV_G→GMV_H 07:54–07:56; wait GMV_H 07:56–08:00
V7	Partille centrum · PC_7, PC_9	08:27	08:27	same reference gap PC_7→PC_9 08:27–08:27
V8	Jons väg · JON_B, JON_A	08:42	08:45	deadhead JON_B→JON_A 08:42–08:43; wait JON_A 08:43–08:45
V9	Partille centrum · PC_7	08:57	08:57	Immediate departure
V10	Jons väg · JON_B, JON_A	09:14	09:25	deadhead JON_B→JON_A 09:14–09:15; C2 charge JON_A 09:15–09:25 +44.832kWh
V11	Partille centrum · PC_7, PC_5	09:37	09:37	same reference gap PC_7→PC_5 09:37–09:37
V12	Heden · HED_E, HED_D	10:08	10:13	deadhead HED_E→HED_D 10:08–10:10; wait HED_D 10:10–10:13
V13	Partille centrum · PC_5	10:37	10:37	Immediate departure
V14	PARX depot · PARX	10:42	12:43	prep PARX 10:42–10:45; C3 charge PARX 10:45–12:30 +105.000kWh; wait PARX 12:30–12:36; prep PARX 12:36–12:43
V15	Jons väg · JON_A	12:54	12:54	Immediate departure
V16	Partille centrum · PC_9, PC_7	13:07	13:10	same reference gap PC_9→PC_7 13:07–13:10
V17	Gamlestads Torg · GMV_G, GMV_H	13:34	13:42	deadhead GMV_G→GMV_H 13:34–13:36; wait GMV_H 13:36–13:42
V18	Partille centrum · PC_7	14:07	14:10	wait PC_7 14:07–14:10
V19	Gamlestads Torg · GMV_G, GMV_H	14:35	14:42	deadhead GMV_G→GMV_H 14:35–14:37; wait GMV_H 14:37–14:42
V20	Partille centrum · PC_7	15:07	15:10	wait PC_7 15:07–15:10
V21	Gamlestads Torg · GMV_G, GMV_H	15:35	15:40	deadhead GMV_G→GMV_H 15:35–15:37; wait GMV_H 15:37–15:40
V22	Partille centrum · PC_7	16:07	16:07	Immediate departure
V23	Heden · 3127L, HED_D	16:22	16:50	C4 charge 3127L 16:22–16:49 +130.728kWh; deadhead 3127L→HED_D 16:49–16:50
V24	Partille centrum · PC_5	17:17	17:17	Immediate departure
V25	Heden · HED_E, HED_D	17:48	17:52	deadhead HED_E→HED_D 17:48–17:50; wait HED_D 17:50–17:52
V26	Partille centrum · PC_5	18:17	18:17	Immediate departure
V27	Heden · HED_E	18:47	18:47	Immediate departure
V28	PARX depot · PARX	19:02	19:05	prep PARX 19:02–19:05

Duty 13309 — complete event ledger

Includes preparation, recorded platform movements, waits and gaps between platform codes without an explicit movement record. Such gaps are not assigned an invented travel time or energy. Source rows refer to Par_VehicleDetails.xlsx, Data worksheet.

Event	Graph	Kind	Recorded endpoints	Start	End	Trip	Recharge kWh	Source row
E1		prep	PARX → PARX	05:12	05:19			A543:X543
E2	M1	deadhead	PARX → HED_D	05:19	05:37			A544:X544
E3	L1	service	HED_D → PC_10	05:37	06:07	226		A545:X545
E4	L2	service	PC_10 → HED_E	06:07	06:41	229		A546:X546
E5		deadhead	HED_E → 3127L	06:41	06:42			A547:X547
E6	C1	charge	3127L → 3127L	06:42	06:56		40.594	A548:X548
E7		deadhead	3127L → HED_D	06:56	06:57			A549:X549
E8	L3	service	HED_D → PC_10	06:57	07:27	234		A550:X550
E9		same reference gap	PC_10 → PC_7	07:27	07:30			rows 550–551
E10	L4	service	PC_7 → GMV_G	07:30	07:54	586		A551:X551
E11		deadhead	GMV_G → GMV_H	07:54	07:56			A552:X552
E12		wait	GMV_H → GMV_H	07:56	08:00			rows 552–553
E13	L5	service	GMV_H → PC_7	08:00	08:27	589		A553:X553
E14		same reference gap	PC_7 → PC_9	08:27	08:27			rows 553–554
E15	L6	service	PC_9 → JON_B	08:27	08:42	901		A554:X554
E16		deadhead	JON_B → JON_A	08:42	08:43			A555:X555
E17		wait	JON_A → JON_A	08:43	08:45			rows 555–556
E18	L7	service	JON_A → PC_7	08:45	08:57	792		A556:X556
E19	L8	service	PC_7 → JON_B	08:57	09:14	793		A557:X557
E20		deadhead	JON_B → JON_A	09:14	09:15			A558:X558
E21	C2	charge	JON_A → JON_A	09:15	09:25		44.832	A559:X559
E22	L9	service	JON_A → PC_7	09:25	09:37	796		A560:X560
E23		same reference gap	PC_7 → PC_5	09:37	09:37			rows 560–561
E24	L10	service	PC_5 → HED_E	09:37	10:08	393		A561:X561
E25		deadhead	HED_E → HED_D	10:08	10:10			A562:X562
E26		wait	HED_D → HED_D	10:10	10:13			rows 562–563
E27	L11	service	HED_D → PC_5	10:13	10:37	396		A563:X563
E28	M2	deadhead	PC_5 → PARX	10:37	10:42			A564:X564
E29		prep	PARX → PARX	10:42	10:45			A565:X565
E30	C3	charge	PARX → PARX	10:45	12:30		105.000	A566:X566
E31		wait	PARX → PARX	12:30	12:36			rows 566–567

Event	Graph	Kind	Recorded endpoints	Start	End	Trip	Recharge kWh	Source row
E32		prep	PARX → PARX	12:36	12:43			A567:X567
E33	M3	deadhead	PARX → JON_A	12:43	12:54			A568:X568
E34	L12	service	JON_A → PC_9	12:54	13:07	922		A569:X569
E35		same reference gap	PC_9 → PC_7	13:07	13:10			rows 569–570
E36	L13	service	PC_7 → GMV_G	13:10	13:34	626		A570:X570
E37		deadhead	GMV_G → GMV_H	13:34	13:36			A571:X571
E38		wait	GMV_H → GMV_H	13:36	13:42			rows 571–572
E39	L14	service	GMV_H → PC_7	13:42	14:07	629		A572:X572
E40		wait	PC_7 → PC_7	14:07	14:10			rows 572–573
E41	L15	service	PC_7 → GMV_G	14:10	14:35	632		A573:X573
E42		deadhead	GMV_G → GMV_H	14:35	14:37			A574:X574
E43		wait	GMV_H → GMV_H	14:37	14:42			rows 574–575
E44	L16	service	GMV_H → PC_7	14:42	15:07	635		A575:X575
E45		wait	PC_7 → PC_7	15:07	15:10			rows 575–576
E46	L17	service	PC_7 → GMV_G	15:10	15:35	638		A576:X576
E47		deadhead	GMV_G → GMV_H	15:35	15:37			A577:X577
E48		wait	GMV_H → GMV_H	15:37	15:40			rows 577–578
E49	L18	service	GMV_H → PC_7	15:40	16:07	641		A578:X578
E50	M4	deadhead	PC_7 → 3127L	16:07	16:22			A579:X579
E51	C4	charge	3127L → 3127L	16:22	16:49		130.728	A580:X580
E52		deadhead	3127L → HED_D	16:49	16:50			A581:X581
E53	L19	service	HED_D → PC_5	16:50	17:17	447		A582:X582
E54	L20	service	PC_5 → HED_E	17:17	17:48	451		A583:X583
E55		deadhead	HED_E → HED_D	17:48	17:50			A584:X584
E56		wait	HED_D → HED_D	17:50	17:52			rows 584–585
E57	L21	service	HED_D → PC_5	17:52	18:17	456		A585:X585
E58	L22	service	PC_5 → HED_E	18:17	18:47	459		A586:X586
E59	M5	deadhead	HED_E → PARX	18:47	19:02			A587:X587
E60		prep	PARX → PARX	19:02	19:05			A588:X588

Scope: recorded GIRO original only; no optimized counterpart or new feasibility certificate. The four recharge amounts were independently checked against raw workbook cells and this duty’s 239.01 kWh18E2 capacity. The primary schematic displaces PARX for readability; coordinates.csv retains actual documented proxies and source URLs. Full extraction, source hashes and validation are in schedules.json and validation.json.